
Activity: The Spaza Shop Slap Chips Calculator

You and your friends are hanging out and decide to make a massive batch of **Slap Chips**. You go down to the local Spaza shop to buy loose potatoes. Each potato bag has its weight measured in grams. You need to write a program that calculates the total weight of the potatoes in kilograms, the total cost, and the final cost including **15% VAT**.

Instructions

Complete the program below by following the specific requirements. Ensure your variables are well-named and that you follow standard coding practices.

Questions

1. Write the necessary include headers, the main function structure, and the proper return statement. **[2 Marks - C1]**
2. Declare and initialize **two named constants**: one for the VAT rate (15%) and one for the conversion factor from grams to kilograms (1000). **[2 Marks - C1]**
3. Declare variables for the number of potato bags and the weight per bag in grams. Choose the most **suitable data types and write a brief comment explaining your choice**. **[2 Marks - C2]**
4. Prompt the user to enter the number of potato bags and the weight of one bag in grams, then read the inputs. **[2 Marks - C1]**
5. Calculate the total weight in kilograms **without causing integer division**. *(Hint: 12 bags × 230 grams = 2760g. Dividing by 1000 should give 2.76kg, not 2).* **[3 Marks - C2]**
6. Calculate the total cost (assume a fixed price of R25.00 per kilogram of potatoes) and the final cost including VAT. **[3 Marks - C2]**
7. Display the total weight in kilograms and the final price including VAT. Ensure the output is clearly labelled and formatted to 2 decimal places. **[2 Marks - C2]**
8. Imagine a classmate wrote the weight calculation line like this:

```
double totalKg = (numBags * weightInGrams) / 1000;
```

If a user inputs 12 bags and 230 grams, the program displays 2 kg instead of 2.76 kg.

- **Identify** why this logical error happens.
- **Rewrite** the line of code to fix it permanently . **[4 Marks -]**

TOTAL: 20 MARKS

Marking Rubric

Criterion	Level	Marks	Allocation Guide
Headers, main, return	C1	**2**	1 mark for #include <iostream>, 1 mark for structural setup and return 0;.
Two named constants	C1	**2**	1 mark per properly declared constant using the const keyword.
Suitable types with a reason	C2	**2**	1 mark for correct data types; 1 mark for valid code comment logic.
Prompts and input	C1	**2**	1 mark for clear cout prompts; 1 mark for correct cin syntax.
Kilograms without int division	C2	**3**	1 mark for formula logic; 2 marks for ensuring the operation evaluates using floating-point math.
Cost and VAT included	C2	**3**	1.5 marks for base cost calculations; 1.5 marks for correctly calculating/adding the 15% VAT.
Formatted, labelled output	C2	**2**	1 mark for user-friendly labels; 1 mark for implementing fixed and setprecision(2). *(Allows learners who missed formatting to still gain full arithmetic marks).*
Find and fix the int error (C3)	C3	**4**	**Reasoning (2 marks):** Explaining that numBags and weightInGrams are integers, causing integer division to drop the decimal .76 before storing it in a double. **Fix (2 marks):** Changing the literal divider to 1000.0, using a

		double constant, or implementing an explicit type cast like static_cast<double>().
Total	**20**	

```
// Criteria 1: Headers, main, return [2 Marks]
#include <iostream>
#include <iomanip> // For formatting output

using namespace std;

int main() {
    // Criteria 2: Two named constants [2 Marks]
    const double VAT_RATE = 0.15;
    const double GRAMS_TO_KG = 1000.0; // Must be double to prevent int division
    const double PRICE_PER_KG = 25.00;

    // Criteria 3: Suitable types with a reason [2 Marks]
    int numBags;         // int: because you buy a whole number of potato bags
    double weightInGrams; // double: allows for precise precision in weight measurements

    // Calculation variables
    double totalKg, rawCost, totalCostWithVat;

    // Criteria 4: Prompts and input [2 Marks]
    cout << "Enter the number of potato bags: ";
    cin >> numBags;
    cout << "Enter the weight of one bag (in grams): ";
    cin >> weightInGrams;
```

```
    // Criteria 5: Kilograms without int division [3 Marks]
    totalKg = (numBags * weightInGrams) / GRAMS_TO_KG;

    // Criteria 6: Cost and VAT included [3 Marks]
    rawCost = totalKg * PRICE_PER_KG;
    totalCostWithVat = rawCost + (rawCost * VAT_RATE);

    // Criteria 7: Formatted, labelled output [2 Marks]
    cout << fixed << setprecision(2);
    cout << "\n--- Slap Chips Order Summary ---" << endl;
    cout << "Total potato weight: " << totalKg << " kg" << endl;
    cout << "Total cost (incl. VAT): R" << totalCostWithVat << endl;

    return 0;
}
```